

USEME - HW6 - Image Processing

How to Run The Program

Some options for users to choose

The users still can use everything from Usume in homework 5 with some more options from this homework 6

1/ User could run the GuiMain, it will show the interactive view window. Thus, this is very user-friendly so I think everyone would know how to use it when looking at it. This is the instructions:

- Users could click the load and select the image they want to load
- Users could do all the effects that include in the view, and they will see the current image they are working on with. If they change the effects, the image will update automatically.
- Users could see the histogram of the red, green, and blue distribution and intensity simply by clicking the red histogram, green histogram, blue histogram, and intensity buttons. After using the effects, if the users want to look at the new histogram, they have to press the button again.
- Users could save a file by press the save button and enter the name of the image as well as the type of image. For example, abcd.jpg

2/ User could run in the terminal, they have to type
cd res

After that, they could type: java -jar ImageProcessing.jar

This will open the graphic user interface and do everything I said previous

Or else, while still in cd res users could type java -jar ImageProcessing.jar -file script2.txt

This will be the same in the USEME in hw5, it will run the script2.txt and user could create new text file of command if they want to

Or else, while still in cs res, users could type java -jar ImageProcessing.jar -text

This will show the interactive text mode, which is the same in USEME for hw5, users could do every functionalities such as

1/ Load command (could load in different file such as ppm, jpg, png, bmp)

#load image.ppm and call it img

load res/image.ppm img

#if the users also have different file formats (for example image.png), they can simply write dote and file format after the image. For example when they want to load a file png:

load res/image.png img

(other options are .jpg and .bmp)

2/ Save command for ppm

#after we load the img, we can save the img

save testsave.ppm img

#if the users also have different file formats (for example image.png), they can simply write dote and file format after the image. For example when they want to save a file png:

save res/image.png img

(other options are .jpg and .bmp)

3/ Flip vertical

flip the img and call it img-vertical

vertical-flip img img-vertical

4/ Flip horizontal

flip the img and call it img-horizontal

horizontal-flip img img-horizontal

5/ Brighten

#brighten the img by 50 levels and call it img-brighten

brighten 50 img img-brighten

6/ Value-Component

#use value-component effect and call it img-value

value-component img img-value

7/ Intensity-Component

#use intensity-component effect and call it img-intensity

intensity-component img img-intensity

8/ Luma-Component

#use luma-component effect and call it img-luma

luma-component img img-luma

9/ Red-Component

#use red-component effect and call it img-red

red-component img img-red

10/ Green-Component

#use green-component effect and call it img-green

green-component img img-green

11/ Blue-Component

#use blue-component effect and call it img-blue

blue-component img img-blue

12/ Blur

use blur effect and call it bl

blur img bl

13/ Sharpen

use sharpen effect and call it sh

sharpening img sh

14/ Greyscale

use greyscale effect and call it grey

greyscale img grey

15/ Sepia

use sepia effect and call it sep

sepia img sep

One reminder, if the user do not do cd res, they have to make sure to call out the res folder of the images as well as the jar file because if not the app will have no idea where to find the images and the jar.